

- 3. Integers:** All counting numbers, zero and negatives of counting numbers, form the set of integers.
Thus,, $-3, -2, -1, 0, 1, 2, 3, \dots$ are all integers.
Set of positive integers = $\{1, 2, 3, 4, 5, 6, \dots\}$
Set of negative integers = $\{-1, -2, -3, -4, -5, -6, \dots\}$
Set of all non-negative integers = $\{0, 1, 2, 3, 4, 5, \dots\}$
- 4. Even Numbers:** A counting number divisible by 2 is called an even number.
Thus, $0, 2, 4, 6, 8, 10, 12, \dots$ etc. are all even numbers.
- 5. Odd Numbers:** A counting number not divisible by 2 is called an odd number.
Thus, $1, 3, 5, 7, 9, 11, 13, \dots$ etc. are all odd numbers.
- 6. Prime Numbers:** A counting number is called a prime number if it has exactly two factors, namely itself and 1.
Ex. All prime numbers less than 100 are:
 $2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97$
- 7. Composite Numbers:** All counting numbers, which are not prime, are called composite numbers.
A composite number has more than 2 factors.
- 8. Perfect Numbers:** A number, the sum of whose factors (except the number itself), is equal to the number, is called a perfect number, e.g. 6, 28, 496
The factors of 6 are 1, 2, 3 and 6. And, $1 + 2 + 3 = 6$.
The factors of 28 are 1, 2, 4, 7, 14 and 28. And, $1 + 2 + 4 + 7 + 14 = 28$.
- 9. Co-primes (or Relative Primes):** Two numbers whose H.C.F. is 1 are called co-prime numbers,
Ex. (2, 3), (8, 9) are pairs of co-primes.
- 10. Twin Primes:** Two prime numbers whose difference is 2 are called twin-primes,
Ex. (3, 5), (5, 7), (11, 13) are pairs of twin-primes.
- 11. Rational Numbers:** Numbers which can be expressed in the form $\frac{p}{q}$, where p and q are integers and $q \neq 0$, are called rational numbers.
Ex. $\frac{1}{8}, \frac{-8}{11}, 0, 6, 5\frac{2}{3}$ etc.
- 12. Irrational Numbers:** Numbers which when expressed in decimal would be in non-terminating and non-repeating form, are called irrational numbers.
Ex. $\sqrt{2}, \sqrt{3}, \sqrt{5}, \sqrt{7}, \pi, e, 0.231764735\dots$

IV. Important Facts:

- All natural numbers are whole numbers.
- All whole numbers are not natural numbers.
0 is a whole number which is not a natural number.
- Even number + Even number = Even number
Odd number + Odd number = Even number
Even number + Odd number = Odd number
Even number - Even number = Even number
Odd number - Odd number = Even number
Even number - Odd number = Odd number
Odd number - Even number = Odd number
Even number $\times$ Even number = Even number
Odd number $\times$ Odd number = Odd number
Even number $\times$ Odd number = Even number
- The smallest prime number is 2.
- The only even prime number is 2.
- The first odd prime number is 3.
- 1 is a unique number - neither prime nor composite.
- The least composite number is 4.
- The least odd composite number is 9.
- Test for a Number to be Prime:**
Let p be a given number and let n be the smallest counting number such that $n^2 \geq p$.
Now, test whether p is divisible by any of the prime numbers less than or equal to n . If yes, then p is not prime otherwise, p is prime.